

TOMAS BENCOMO

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tjbencomo.github.io

EDUCATION

MD/PhD	University of Washington Medical Scientist Training Program Mentors: Paul Nghiem, Manu Setty, Evan Newell	2022 - present
BS	Stanford University, Computer Science Concentration in Biocomputation GPA: 3.99	2017 - 2021

HONORS AND AWARDS

Poster Prize – Fred Hutch Basic Sciences Division Retreat Third place award for best poster	2025
Oliver W. Press ARCS Endowment Award Achievement Reward For College Scientist scholarship to help fund PhD training	2024
Stanford Bio-X Summer Research Grant Funding for undergraduate research in the Dermatology department	2020
Tau Beta Pi Member Engineering honors society at Stanford	2020
Stanford VPUE Conference Grant Award to present research at Society For Investigative Dermatology Annual Meeting	2020
Mayo Clinic Summer Undergraduate Research Fellowship Fellow studying glioblastoma in the Precision NeuroTherapeutics Program	2018 & 2019
TGen Helios Scholar High school intern in the Helios Program at TGen	2016
Helios Symposium Poster Award Winner Awarded for a top 10 poster at the intern symposium	2016

RESEARCH EXPERIENCE

Graduate Student , University of Washington & Fred Hutch Cancer Center Advisors: Paul Nghiem MD/PhD, Manu Setty PhD, and Evan Newell PhD - Studied mechanisms of immunotherapy resistance using spatial multi-omics technologies - Developed novel computational approaches for spatial transcriptomics data analysis	2023 – present
Research Associate , Lee Lab, Stanford Medicine Advisor: Carolyn Lee MD/PhD	2021 – 2022

- Investigated mechanisms of perineural invasion using single cell sequencing
- Performed a meta-analysis of publicly available cSCC RNA-Seq datasets
- Studied metabolic reprogramming caused by non-coding mutations in OXA1L

Undergraduate Researcher, Lee Lab, Stanford Medicine

2017 – 2021

Advisor: Carolyn Lee MD/PhD

- Characterized novel genes implicated in skin cancers using multi-omics data
- Examined survival data to find biomarkers for squamous cell carcinoma (SCC)
- Analyzed proteomics data to identify novel interactors for melanoma oncogene RAC1

Summer Undergraduate Research Fellow, Swanson Lab, Mayo Clinic

2018-2019

Advisor: Kristin Swanson PhD

- Analyzed MRI data to study blood brain barrier disruption in glioblastoma (GBM)
- Trained machine learning models to predict drug distribution in the brain from MRI
- Built prognostic models using radiomics data to predict patient survival

Helios Scholar, Translational Genomics Research Institute

2016

Advisor: Seungchan Kim PhD and Gil Speyer PhD

- Redesigned the lab's EDDY algorithm to use NVIDIA's CUDA API
- Helped secure \$200,000 Compute 4 the Cure grant from NVIDIA
- Achieved 500x speedup; the algorithm can now analyze previously impossible datasets

PUBLICATIONS ([GOOGLE SCHOLAR](#))

Articles

Gene expression landscape of cutaneous squamous cell carcinoma progression 2024

Tomas Bencomo, Carolyn S Lee

British Journal of Dermatology

Unravelling the landscape of skin cancer through single-cell transcriptomics 2023

Ankit Srivastava, **Tomas Bencomo**, Ishani Das, Carolyn S. Lee

Translational Oncology

MAB21L4 deficiency drives squamous cell carcinoma via activation of RET 2022

Ankit Srivastava, Cristina Tommasi, Dane Sessions, Angela Mah, **Tomas Bencomo**, Jasmine M Garcia, Tiffany Jiang, Michael Lee, Joseph Y Shen, Lek Wei Seow, Audrey Nguyen, Kimal Rajapakshe, Cristian Coarfa, Kenneth Y Tsai, Vanessa Lopez-Pajares, Carolyn S Lee

Cancer Research

Mutant collagen COL11A1 enhances cancerous invasion 2021

Carolyn Lee, Zurab Siprashvili, Angela Mah, **Tomas Bencomo**, Lara Elcavage, Yonglu Che, Rajani Shenoy, Sumaira Aasi, Paul Khavari

Oncogene

Preprints

ecDNA in cutaneous squamous cell carcinoma is associated with increased nodal disease 2024

Tomas Bencomo, Amarinder Thind, Bruce Ashford, Marie Ranson, Carolyn S. Lee

Biorxiv

Conference Abstracts

Gene expression landscape of cutaneous squamous cell carcinoma progression

T. Bencomo, C. S. Lee

Society For Investigative Dermatology 2024 Annual Meeting

ecDNA in cutaneous squamous cell carcinoma is associated with increased nodal disease

T. Bencomo, A. Thind, I. Donohue, B. Ashford, M. Ranson, C. S. Lee

Society For Investigative Dermatology 2024 Annual Meeting

Mast cells orchestrate perineural invasion in aggressive epithelial cancers

A. Srivastava , **T. Bencomo** , A. Mah , J. Garcia , I. Donohue , A. Tan , L. Seow , A. Nguyen , T. Jiang , P. Dwivedi , C. M. Rose , R. Brown , S. Aasi , C. S. Lee

Society For Investigative Dermatology 2024 Annual Meeting

Metabolic reprogramming by cancer-associated mutant OXA1L promotes tumor progression and dedifferentiation

A. Srivastava, I. Donohue, A. Tan, A. Nguyen, **T. Bencomo**, J. Ye, B. Ashford, M. Ranson, D. Kashatus, C. S. Lee

Society For Investigative Dermatology 2024 Annual Meeting

Chromosomal structural variants but not extrachromosomal DNAs are common in Merkel cell carcinoma

J. D. Bloomstein , J. Luebeck , J. Rose , **T. Bencomo**, D. S. Hippe , S. Jani, V. Bafna , H. Chang , P. Nghiem

Society For Investigative Dermatology 2024 Annual Meeting

Somatic Mutation of the OXA1L 5'UTR enables Cutaneous Squamous Cell Carcinoma

Angela Mah, Jasmine Garcia, Dane Sessions, **Tomas Bencomo**, Ashley Amado, Ankit Srivastava, Carolyn Lee

Plenary Session - Society For Investigative Dermatology 2021 Annual Meeting

Molecular Profiling of Cutaneous C-Group Non-Langerhans Cell Histiocytoses

Rebekah Wieland, **Tomas Bencomo**, Carolyn Lee, RYANNE BROWN

United States and Canadian Academy of Pathology 2021 Annual Meeting

Sex Differences In GBM Patient Survival As A Function of Extent Of Surgical Resection and Cycles of Adjuvant Temozolomide During Standard Of Care Regimens

Julia Lorence, **Tomas Bencomo**, Haylye White, Cassandra Rickertsen, Susan Massey, Kyle Singleton, Andrea Hawkins-Daarud, Sandra Johnston, Alyx Porter, Maciej Mrugala, Bernard Bendok, Leland Hu, Joshua Rubin, Kristin Swanson

Society for Neuro-Oncology 2020 Annual Meeting

Rac1-interacting proteins are prognostic factors for melanoma survival

Marten CG Winge, **Tomas Bencomo**, M. Peter Marinkovich, Carolyn S. Lee

Society For Investigative Dermatology 2020 Annual Meeting

Sex differences in GBM treatment: an observational study

Lorence J., **Bencomo T.**, White H., Rickertsen C.R., De Leon, G. Singleton K. W., Daruud-Hawkins A., Bendok B. R., Porter A. B., Mrugala M. M., Rubin J.B., and Swanson K. R.

3rd Annual Celebration of Women's Health Research – Sexx as a Biological Variable

Sex Differences in Tumor Location Incidence in Newly Diagnosed Glioblastoma Patients

Lorence J., **Bencomo T.**, White H., Rickertsen C.R., De Leon G., Singleton K., Bendok B., Porter A. B., Rubin J. B., Swanson K. R.

Society for Neuro-Oncology 2019 Annual Meeting

Modeling the interface between non-invasive imaging and drug distribution

Ranjbar, Sara; **Bencomo, Tomas**; Jackson, Pamela R; Randall, Elizabeth; Regan, Micheal; Abdelmoula, Walid M; Lopez, Begona GC; Massey, Susan Christine; He, Lihong; Macura, Slobodan; Hu, Leland; Agar, Jefferey N; Sarkaria, Jann; Agar, Nathalie; Swanson, Kristin R.

CSBC/PS-ON 2019 Annual Investigators Meeting

The Same But Different: Identifying Distinct Imaging Ecologies in Male and Female Glioblastomas

Tomas Bencomo, Andrea Hawkins-Daarud, Kyle Singleton and Kristin R. Swanson.

CSBC/PS-ON 2018 Annual Investigators Meeting

GPU-accelerated differential dependency network analysis

Gil Speyer, Juan Rodriguez, **Tomas Bencomo** and Seungchan Kim

Euromicro International Conference on Parallel, Distributed and Network-Based Processing 2018

TALKS

Spatial interaction between tumor cells and immune hotspots defines MCC ICI resistance	2026
4 th International Merkel Cell Carcinoma Symposium – Invited Speaker	
Spatial transcriptomics analysis of MCC reveals factors involved in T cell exclusion	2025
9 th Nordic/2 nd European Merkel Cell Carcinoma Meeting	
Spatial transcriptomics analysis of immunotherapy response in Merkel cell carcinoma	2025
Society for Investigative Dermatology Annual Meeting	
Spatial transcriptomics using Xenium: An Exciting New Frontier	2024
Japanese Society of Investigative Dermatology Annual Meeting – Invited Speaker	
Spatial transcriptomics analysis of MCC reveals difficult to phenotype immune cell subsets	2024
10x Seattle Single Cell and Spatial Symposium	
Characterizing the tumor microenvironment of MCC with spatial transcriptomics	2024
Seattle Translational Tumor Research Skin Oncology	
Joint modeling of cellular morphology and gene expression at single-cell resolution	2024
Fred Hutch Computational Biology Retreat: Trainee Elevator Pitch Presentation	

LANGUAGES

English: Native Language

Spanish: Intermediate Proficiency

TECHNICAL SKILLS

Programming: R, Python, Java, C, C++

Biostatistics: Regression modeling, machine learning, survival analysis, experimental design

Bioinformatics: Mutation profiling, bulk and single cell RNA-sequencing, spatial analysis, data visualization

Computing: Snakemake, Docker, Unix, Bash, HPC, scikit-learn, pandas, tidyverse

EXTRACURRICULARS

Tau Beta Pi Honors Society Board – Academic Co-Chair	2020 – 2021
Organize academic development events and lead the Peer Mentorship Program	

Computer Science Peer Mentor	2020 – 2021
Advise and mentor underclassmen considering a major in Computer Science	